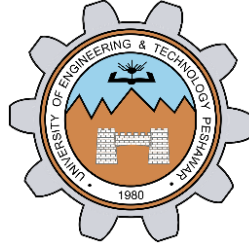


Investigate the network using “ping”, “tracert”

LAB # 02



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CSE-303L Data Communication & Networks Lab

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“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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Date:

22nd March 2024

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Lab Objectives:

- Learn to use the TCP/IP Packet Internet Groper (ping) command.
- Learn to use the Trace Route (tracert) command.

About Ping:

The original PING command stood for "Packet Internet Groper", and was a package of diagnostic utilities used by DARPA personnel to test the performance of the ARPANET. However, the modern Internet Ping command refers to a program written by Mike Muss in December, 1983, which has since become one of the most versatile and widely used diagnostic tools on the Internet.

Typical Diagnostic Tests Performed by Ping Command

Some of the internet diagnostic tests performed by ping command are:

- **Access** – Ping is used to determine whether the remote host is active or inactive. If a certain site is not pinged, but the other sites can, then it's a pretty good sign that your Internet network is fine and that site is down. On the other hand, if you can't ping any site, then likely your entire network connection is down that needs rebooting.
- **Time & distance** – Another use of Ping command is to determine how long it takes to bounce a packet off of another site. Thereby giving Internet distance in network terms. For example, a web site hosted on your neighbor's computer with a different Internet service provider (ISP) might go through more routers and be farther away in network distance than a site on the other side of the ocean with a direct connection to the Internet backbone. If a site seems slow, then ping distance of that site can be compared with that of other Internet sites to find out whether it is the site, the network, or your system that is slow. You can also compare ping times to get an idea of which sites have the fastest network access and would be most efficient for downloading, chatting, and other applications.
- **Domain IP address** – Typically, Ping command is used to probe either a domain name or an IP address; if a domain name is pinged, and then it displays the corresponding IP address in its response.

Usage

ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS] [-r count] [-s count] [[-j host-list] | [-k host-list]] [-w timeout] target_name

where target_name can be either IP Address or host name.

Option Description

=====

-t Ping the specified host until stopped. To see stats & Continue, type Control-Break; to stop, type

Control-C

-a (Resolve Addresses) Resolve address to host names

-n count (Echo Count) Sends echo packets specified by count; **Default is 4**

-l size (Send buffer Size) Use to increase or decrease the size of the ICMP packets sent in the ping request;

Min=0, Max=65500

-i TTL (Time to Live) set the number of hops (routers) that the ping request can traverse before it is discarded. The TTL field in the packet is decremented by one each time it passes through a router. When the number reaches zero, the router discards the packet and sends a TTL Expired ICMP message back.

-w timeout Timeout in seconds that ping waits for each reply

=====

Examples

Ping a Local host

In this example, a local host is pinged to see if the computer is able to send and receive information.

Step 1

Use the Start menu to open the command-prompt:

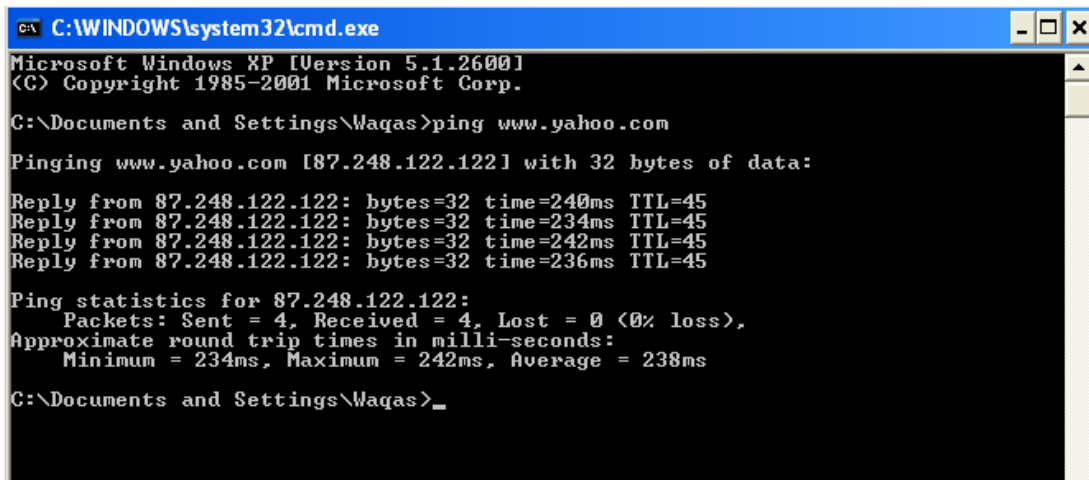
Start>Programs>Accessories>Command Prompt or

Start>Programs>Command Prompt

Step 2

Type **ping, space,** and **192.168.2.x.** Press Enter key.

Since Ping uses the **Internet Control Message Protocol (ICMP)** echo-request and echo-reply feature to test physical connectivity, it reports on four attempts and gives an indication the reliability of the connection. Look over the result and verify that the ping was successful. Was the ping successful? A successful host ping is shown in Figure 2.1.



```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Waqas>ping www.yahoo.com

Pinging www.yahoo.com [87.248.122.122] with 32 bytes of data:

Reply from 87.248.122.122: bytes=32 time=240ms TTL=45
Reply from 87.248.122.122: bytes=32 time=234ms TTL=45
Reply from 87.248.122.122: bytes=32 time=242ms TTL=45
Reply from 87.248.122.122: bytes=32 time=236ms TTL=45

Ping statistics for 87.248.122.122:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 234ms, Maximum = 242ms, Average = 238ms

C:\Documents and Settings\Waqas>_
```

Figure 2.1: Command Screen for ping www.yahoo.com

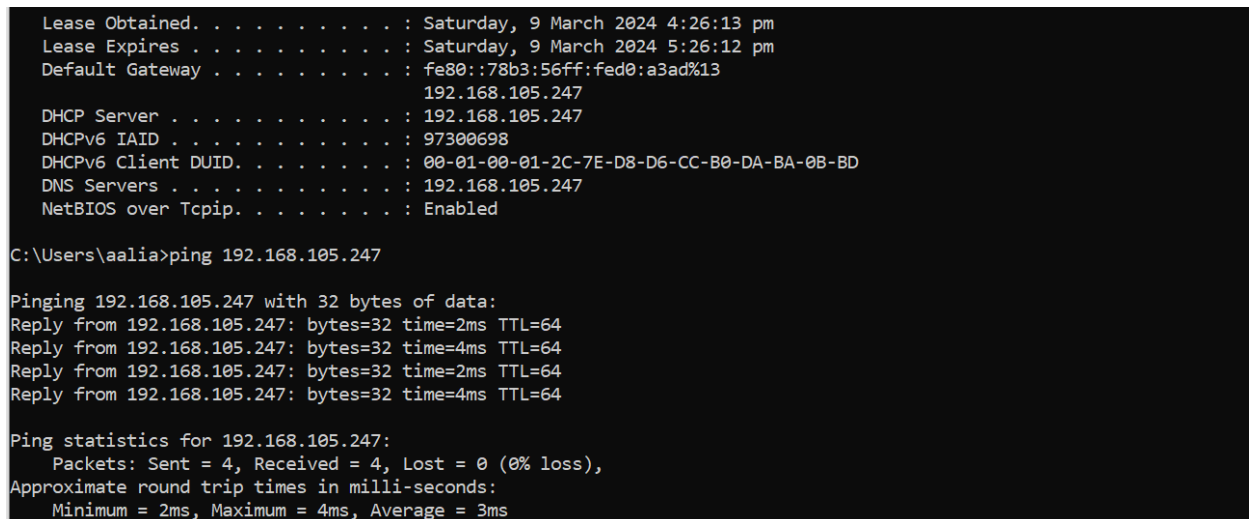
The first output line shows the Fully Qualified Domain Name (FQDN) followed by the IP address.

A Domain Name Service (DNS) server somewhere in the network was able to resolve the name to an IP address. DNS servers resolve domain names, not hostnames, to IP addresses. Without this name resolution, the ping would have failed because TCP/IP only understands valid IP addresses. It would not be possible to use the web browser without this name resolution. With DNS, connectivity to computers on the Internet can be verified using a familiar web address, or domain name, without having to know the actual IP address. If the nearest DNS server does not know the IP address, the server asks a DNS server higher in the Internet structure.

-----TASK 01-----

a) Ping the IP address of the Default Gateway and DNS Servers. Was the result successful?

Ans: Yes. The result was successful as seen in the screenshot below.



```
Lease Obtained. . . . . : Saturday, 9 March 2024 4:26:13 pm
Lease Expires . . . . . : Saturday, 9 March 2024 5:26:12 pm
Default Gateway . . . . . : fe80::78b3:56ff:fed0:a3ad%13
                           192.168.105.247
DHCP Server . . . . . : 192.168.105.247
DHCPv6 IAID . . . . . : 97300698
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-7E-D8-D6-CC-B0-DA-BA-0B-BD
DNS Servers . . . . . : 192.168.105.247
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\aalialia>ping 192.168.105.247

Pinging 192.168.105.247 with 32 bytes of data:
Reply from 192.168.105.247: bytes=32 time=2ms TTL=64
Reply from 192.168.105.247: bytes=32 time=4ms TTL=64
Reply from 192.168.105.247: bytes=32 time=2ms TTL=64
Reply from 192.168.105.247: bytes=32 time=4ms TTL=64

Ping statistics for 192.168.105.247:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

b) Ping the computer's loop-back address. Type the following command: >> **ping 127.0.0.1**
Ans:

```
C:\Users\aalialia>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

c) What is the IP Address of www.yahoo.com: 2001:4998:18:800::4002: How much time did our ping took to reach www.yahoo.com: 357 ms

```
C:\Users\aalialia>ping www.yahoo.com

Pinging me-ycpi-cf-www.g06.yahoodns.net [2001:4998:18:800::4002] with 32 bytes of data:
Reply from 2001:4998:18:800::4002: time=367ms
Reply from 2001:4998:18:800::4002: time=359ms
Reply from 2001:4998:18:800::4002: time=367ms
Reply from 2001:4998:18:800::4002: time=337ms

Ping statistics for 2001:4998:18:800::4002:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 337ms, Maximum = 367ms, Average = 357ms

C:\Users\aalialia>
```

d) Ping the hostname of another computer. Try to ping the hostname of the computer that was recorded in the previous lab.

```
C:\Users\aalialia>ping DESKTOP-5AU4H87

Pinging DESKTOP-5AU4H87 [fe80::8b66:5439:c659:bd05%12] with 32 bytes of data:
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms

Ping statistics for fe80::8b66:5439:c659:bd05%12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\aalialia>
```

e) Ping the hostname of another computer using `-t`. Try to ping repetitively, the hostname of the computer

```
C:\Users\aalialia>ping DESKTOP-5AU4H87 -t

Pinging DESKTOP-5AU4H87 [fe80::8b66:5439:c659:bd05%12] with 32 bytes of data:
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms
Reply from fe80::8b66:5439:c659:bd05%12: time<1ms

Ping statistics for fe80::8b66:5439:c659:bd05%12:
    Packets: Sent = 7, Received = 7, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
^C
C:\Users\aalialia>
```

f) How can we stop the ping?

Ans: By Pressing Control^C

h) ping the IP address of a DHCP or DNS server

```
C:\Users\aalialia>ping 192.168.105.247

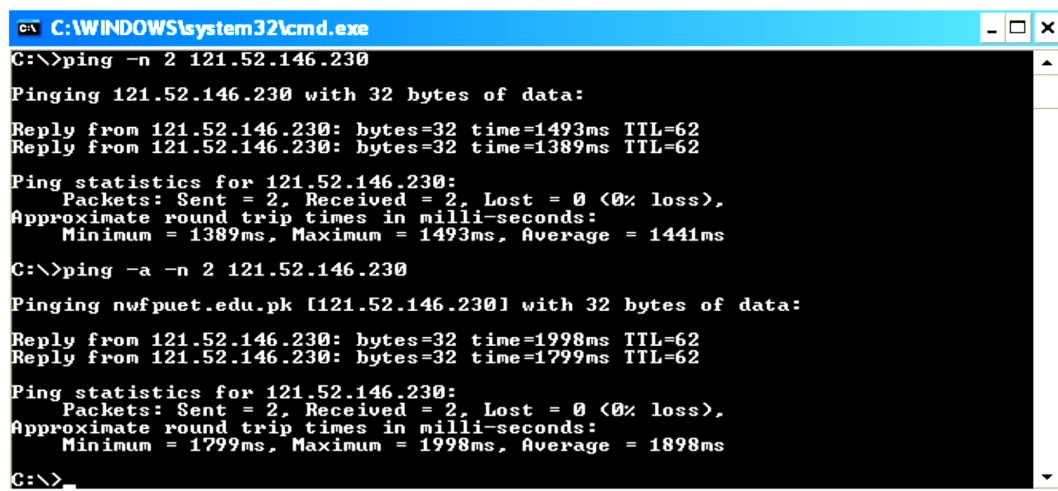
Pinging 192.168.105.247 with 32 bytes of data:
Reply from 192.168.105.247: bytes=32 time=3ms TTL=64
Reply from 192.168.105.247: bytes=32 time=2ms TTL=64
Reply from 192.168.105.247: bytes=32 time=6ms TTL=64
Reply from 192.168.105.247: bytes=32 time=6ms TTL=64

Ping statistics for 192.168.105.247:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 6ms, Average = 4ms

C:\Users\aalialia>
```

Use of Ping Option: Resolve Address

In this example, a website is pinged using its IP Address. By means of ping option: *Resolve Address (-a)*, this IP Address is resolved into its respective host name. Also the ping option: *Echo Count (-n count)* is used to send only 2 echo packets. The details are shown in Figure 2.2.

A screenshot of a Windows command prompt window titled "C:\WINDOWS\system32\cmd.exe". The window shows the execution of two ping commands. The first command is "ping -n 2 121.52.146.230", which results in two successful replies from 121.52.146.230 with 32 bytes of data, times of 1493ms and 1389ms, and a TTL of 62. The second command is "ping -a -n 2 121.52.146.230", which also results in two successful replies from 121.52.146.230 with 32 bytes of data, times of 1998ms and 1799ms, and a TTL of 62. Both commands include ping statistics showing 2 packets sent, 2 received, 0 lost, and approximate round trip times in milliseconds (Minimum, Maximum, Average).

```
C:\WINDOWS\system32\cmd.exe
C:\>ping -n 2 121.52.146.230

Pinging 121.52.146.230 with 32 bytes of data:

Reply from 121.52.146.230: bytes=32 time=1493ms TTL=62
Reply from 121.52.146.230: bytes=32 time=1389ms TTL=62

Ping statistics for 121.52.146.230:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1389ms, Maximum = 1493ms, Average = 1441ms

C:\>ping -a -n 2 121.52.146.230

Pinging nwfpuet.edu.pk [121.52.146.230] with 32 bytes of data:

Reply from 121.52.146.230: bytes=32 time=1998ms TTL=62
Reply from 121.52.146.230: bytes=32 time=1799ms TTL=62

Ping statistics for 121.52.146.230:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1799ms, Maximum = 1998ms, Average = 1898ms

C:\>
```

Figure 2.2 Use of Address Resolution Ping Option

About tracer

Tracer is TCP/IP abbreviation for trace route. The preceding figure shows the successful result when running tracer from Bavaria in Germany. The first output line shows the FQDN followed by the IP address. Therefore, a DNS server was able to resolve the name to an IP address. Then there are listings of all routers the tracer requests had to pass through to get to the destination. tracer uses the same echo requests and replies as the ping command but in a slightly different way. Observe that tracer actually contacted each router three times. Compare the results to determine the consistency of the route. Notice in the below example that there were relatively long delays after router 11, possibly due to congestion. The main thing is that there seems to be relatively consistent connectivity. Each router represents a point where one network connects to another network and the packet was forwarded through.

Usage

The tracer command is used to visually see a network packet being sent and received and the amount of hops required for that packet to get to its destination. Trace route (tracer) is a command-line utility used to trace the path that an Internet Protocol (IP) packet takes to its destination from a source. Tracer will determine the path taken to a destination. It does this by sending Internet Control Message Protocol (ICMP) Echo Request messages to the destination. When sending traffic to the destination, it will increment the Time to Live (TTL) field values to aid in finding the path taken to that destination address. The path is outlined from this process.

tracer [-d] [-h maximum_hops] [-j host-list] [-w timeout] target_name where target_name can be either IP Address or host name.

Option Description

=====

-d (Do Not Resolve Addresses) Displays the route using numeric addresses only rather than showing both IP address and host names, for faster display.

-h maximum_hops (Max. Hops) Specifies the maximum number of hops to use for tracing; Default is 30

-w timeout Specifies how long to wait for a reply to each Request in seconds;

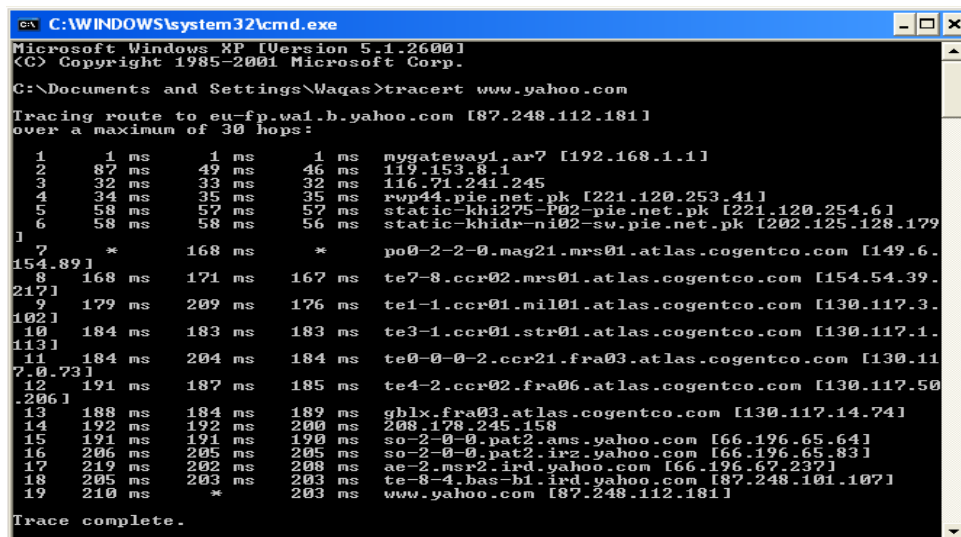
=====

Step 1

Trace the route to the Yahoo web site

Type `tracert www.yahoo.com` and press Enter.

The result shows the complete route to the site, along with the number of hops in the path.



```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Waqas>tracert www.yahoo.com

Tracing route to eu-fp.wai.b.yahoo.com [87.248.112.181]
over a maximum of 30 hops:
  0  1 ms    1 ms    1 ms    mygateway1.ar7 [192.168.1.1]
  1  87 ms   49 ms   46 ms   119.153.8.1
  2  32 ms   33 ms   32 ms   116.71.241.245
  3  34 ms   35 ms   35 ms   rop44.pie.net.pk [221.120.253.41]
  4  58 ms   57 ms   57 ms   static-khi275-P02-pie.net.pk [221.120.254.6]
  5  58 ms   58 ms   56 ms   static-khidr-ni02-sw.pie.net.pk [202.125.128.179]
  6  *        168 ms   *        po0-2-2-0.mag21.mrs01.atlas.cogentco.com [149.6.
154.89]
  7  168 ms  171 ms  167 ms  te7-8.ccr02.mrs01.atlas.cogentco.com [154.54.39.
217]
  8  179 ms  209 ms  176 ms  te1-1.ccr01.mil01.atlas.cogentco.com [130.117.3.
102]
  9  184 ms  183 ms  183 ms  te3-1.ccr01.str01.atlas.cogentco.com [130.117.1.
113]
 10  184 ms  204 ms  184 ms  te0-0-0-2.ccr21.fra03.atlas.cogentco.com [130.11
7.0.73]
 11  191 ms  187 ms  185 ms  te4-2.ccr02.fra06.atlas.cogentco.com [130.117.50
.206]
 12  188 ms  184 ms  189 ms  gblx.fra03.atlas.cogentco.com [130.117.14.74]
 13  192 ms  192 ms  200 ms  208.178.245.158
 14  191 ms  191 ms  190 ms  so-2-0-0.pat2.ams.yahoo.com [66.196.65.64]
 15  206 ms  205 ms  205 ms  so-2-0-0.pat2.irz.yahoo.com [66.196.65.83]
 16  219 ms  202 ms  208 ms  ae-2.msr2.ird.yahoo.com [66.196.67.237]
 17  205 ms  203 ms  203 ms  te-8-4.bas-b1.ird.yahoo.com [87.248.101.107]
 18  210 ms  *        203 ms  www.yahoo.com [87.248.112.181]
 19

Trace complete.
```

Figure 2.3 Command Screen for `tracert www.yahoo.com`

TASK 02

a) Trace the route to the GOOGLE PAKISTAN website by typing:

>> **tracert** www.google.com.pk

```
C:\Users\aalialia> tracert www.google.com.pk

Tracing route to www.google.com.pk [2a00:1450:4019:80c::2003]
over a maximum of 30 hops:

  1    14 ms    18 ms    5 ms    2401:ba80:a103:3a10::c0
  2    *        *        *        Request timed out.
  3    36 ms    32 ms    34 ms    2401:ba80:a0ee::1:1
  4    26 ms    44 ms    38 ms    2401:ba80:a0ee:17::
  5    34 ms    33 ms    54 ms    2401:ba80:a0ee:14::1
  6    35 ms    37 ms    33 ms    2401:ba80:b0ee:5::
  7    *        *        *        Request timed out.
  8    38 ms    35 ms    38 ms    2404:7000:f002:93::1
  9    33 ms    31 ms    36 ms    2404:7000:f001:3::12:22
 10    61 ms    57 ms    60 ms    2404:7000:f001:1::4:18
 11    59 ms    58 ms    52 ms    2404:7000:f001:1::4:6
 12    83 ms    69 ms    *        2001:4860:1:1::b22
 13    69 ms    72 ms    73 ms    2a00:1450:80dc::1
 14    *        *        *        Request timed out.
 15    *        89 ms    82 ms    2001:4860:0:1::8374
 16    80 ms    74 ms    80 ms    2001:4860::9:4002:5a76
 17    77 ms    74 ms    65 ms    2001:4860:0:1::7edf
 18    81 ms    80 ms    76 ms    2001:4860:0:1::52ff
 19    67 ms    72 ms    66 ms    fjr04s08-in-x03.1e100.net [2a00:1450:4019:80c::2003]

Trace complete.

C:\Users\aalialia>
```

The output shows the complete route to the site, along with the number of hops in the path.

b) Trace the route to the UET website using options listed in option description table.

```
C:\Users\aalialia>tracert -h 10 www.uetpeshawar.edu.pk

Tracing route to uetpeshawar.edu.pk [121.52.147.74]
over a maximum of 10 hops:

  1     4 ms     2 ms     5 ms    192.168.105.247
  2     *        *        *        Request timed out.
  3    47 ms    48 ms    80 ms    10.13.89.1
  4    45 ms    35 ms    29 ms    10.12.103.109
  5    28 ms    28 ms    47 ms    172.16.73.29
  6    42 ms    34 ms    43 ms    172.16.73.5
  7     *        *        *        Request timed out.
  8    29 ms    25 ms    21 ms    203.135.56.230
  9    45 ms    36 ms    *        10.15.0.62
 10    75 ms    55 ms    50 ms    202.125.154.162

Trace complete.

C:\Users\aalialia>
```

c) What is the difference between the following commands?

Tracert www.yahoo.com

Tracert -h 20 www.yahoo.com

Ans: First command will trace up to the default maximum number of hops (30). While the second command will only trace up to 20 number of hops because we have explicitly specified it in the command.